

# Min-Yue Chen

Security researcher with 4+ years in vulnerability research, threat analysis, and ML-driven detection for mobile/cellular systems; results adopted by AT&T, recognized by Google, published at ACM MobiCom/MobiSys

Email: [chenmi33@msu.edu](mailto:chenmi33@msu.edu) | [Google Scholar](#) | [Personal Website](#) | 517-920-9648 | East Lansing, MI 48823

## EDUCATION

### Research Area: System Security, Vulnerability Research, AI/ML for Systems & Networks

- **Michigan State University (MSU)** East Lansing, Michigan  
*Ph.D. in Computer Science and Engineering - GPA: 3.95/4* Aug. 2021 - Aug. 2026 (expected)
- **National Chiao Tung University (NCTU)** Hsinchu, Taiwan  
*B.S. in Computer Science* Sep. 2015 - Jun. 2019

## EMPLOYMENT

- **Research Assistant** Aug. 2021 - Present  
Security, Networking, and Mobile Systems Lab, Michigan State University
- **Research Assistant** Jan. 2020 - June 2021  
Networking and Mobile System Lab, National Chiao Tung University

## HONORS AND AWARDS

- ACM MobiSys'26 Featured Paper 2026
- ACM GetMobile Research Highlights 2025, 2023
- ACM SigMobile'24 Research Highlights 2024
- Communications of ACM Research Highlights Nomination 2024
- Google Bug Bounty High-Severity and High-Quality Report 2024
- 911 CVD report contributed to GSMA standard changes 2023
- AT&T Security Award 2023
- ACM MobiCom'22 Best Community Paper Award Runner-Up 2022

## PUBLICATIONS

[MobiSys'26] **Min-Yue Chen**, Y. Hu, Y.-A. Chen, C.-Y. Li, T. Xie, G.-H. Tu, "Insecurity of Lost/Stolen Phone Reporting Services: Vulnerabilities, Attacks, and Countermeasures," *the 24th ACM International Conference on Mobile Systems, Applications, and Services (MobiSys) 2026*, June 21 - 25, 2026, Cambridge, UK. [Acceptance Rate: 68/302 = 22.51%] [Featured Paper](#)

[ToN'26] J. Shi, **Min-Yue Chen**, S. Wang, G.-H. Tu, T. Xie, Y. Hu, M.-H. Chen, H. Yan, C.-Y. Li, C. Peng, "When Mobile Equipment Security Lags Behind Infrastructure: Vulnerabilities, Attacks, and Countermeasures in IMS Services," *accepted by IEEE/ACM Transactions on Networking* [Impact Factor: 5.02]

[GetMobile'25] J. Shi, S. Wang, **Min-Yue Chen**, G.-H. Tu, T. Xie, M.-H. Chen, Y. Hu, C.-Y. Li, and C. Peng, "IMS is Not That Secure on Your 5G/4G Phones," *ACM GetMobile Magazine: Mobile Computing and Communications*, Volume: 29, Issue: 1, pp. 20-24, April, 2025. [Invited Article & Research Highlights](#)

[MobiCom'24] Y. Hu, **Min-Yue Chen**, H. Yan, C.-Y. Cheng, G.-H. Tu, C.-Y. Li, T. Xie, C. Peng, L. Xiao, and J. Tang, "Uncovering Problematic Designs Hindering Ubiquitous Cellular Emergency Services Access," *accepted by ACM the 30th Annual International Conference On Mobile Computing And Networking (MobiCom) 2024*, September 30- October 4, 2024, Washington, D.C., USA. [Acceptance Rate: 55/288 = 19.09%, Winter Run]

- [MobiCom'24] J. Shi, S. Wang, **Min-Yue Chen**, G.-H. Tu, T. Xie, M.-H. Chen, Y. Hu, C.-Y. Li, and C. Peng, "IMS is Not That Secure on Your 5G/4G Phones," accepted by *ACM the 30th Annual International Conference On Mobile Computing And Networking (MobiCom) 2024*, September 30- October 4, 2024, Washington, D.C., USA. [**Acceptance Rate: 48/207 = 23.19%, Summer Run**]
- [ToN'24] **Min-Yue Chen**, Y. Hu, G.-H. Tu, C.-Y. Li, S. Wang, J. Shi, T. Xie, L. Xiao, C. Peng, Z. Tan, and S. Lu, "Taming the Insecurity of Cellular Emergency Services (9-1-1): From Vulnerabilities to Secure Designs," *IEEE/ACM Transactions on Networking*, Volume 32, Issue: 4, pp. 3076-3091, Aug. 2024. [**Impact Factor: 5.02**]
- [ToN'23] S. Wang, T. Xie, **Min-Yue Chen**, G.-H. Tu, C.-Y. Li, X. Lei, P.-Y. Chou, F. Hsieh, Y. Hu, L. Xiao, C. Peng, "Dissecting Operational Cellular IoT Service Security: Attacks and Defenses," *IEEE/ACM Transactions on Networking*, Volume: 32, Issue: 2, pp. 1229-1244, April 2024. [**Impact Factor: 5.02**]
- [CNS'23] J. Shi, T. Xie, G.-H. Tu, C. Peng, C.-Y. Li, A. Hou, S. Wang, Y. Hu, X. Lei, **Min-Yue Chen**, L. Xiao, X. Liu, "When Good Turns Evil: Encrypted 5G/4G Voice Calls Can Leak Your Identities," *IEEE Conference on Communications and Network Security*, October 2- October 5, 2023, Orlando, FL, USA. [**Acceptance Rate: 32/112=28.5%**]
- [GetMobile'23] Y. Hu<sup>(\*)</sup>, **Min-Yue Chen**<sup>(\*)</sup>, G.-H. Tu, C.-Y. Li, S. Wang, J. Shi, T. Xie, L. Xiao, C. Peng, Z. Tan, and S. Lu, "Unveiling the Insecurity of Operational Cellular Emergency Services (911): Vulnerabilities, Attacks, and Countermeasures," *ACM GetMobile Magazine: Mobile Computing and Communications*, Volume: 27, Issue: 1, pp. 39-43, March, 2023. **Invited Article & Research Highlights** (\*): Co-primary author, denotes equal contribution
- [MobiCom'22] Y. Hu<sup>(\*)</sup>, **Min-Yue Chen**<sup>(\*)</sup>, G.-H. Tu, C.-Y. Li, S. Wang, J. Shi, T. Xie, L. Xiao, C. Peng, Z. Tan, and S. Lu, "Uncovering Insecure Designs of Cellular Emergency Services (911)," *ACM the 28th Annual International Conference On Mobile Computing And Networking (MobiCom) 2022*, October 17- October 21, 2022, Sydney, Australia. [**Acceptance Rate: 57/315 = 18.1%**] [**Media Coverage: MSU Today, Wilx TV, Homeland Security News**] **Best Community Paper Award - Runner Up, ACM SIGMOBILE Research Highlights, AT&T Security Award**  
(\*): Co-primary author, denotes equal contribution

## RESEARCH EXPERIENCE

- Research 1. ML-Based Detection of IVR Call Activity** Mar. 2026 - present
- Built an ML pipeline for detecting IVR call states from cellular network traces, including data collection, feature engineering, model training, and evaluation using encrypted MAC/PDCP-layer packet traces; benchmarked *XGBoost*, *1D-CNN*, and *LSTM+Attention* architectures
  - Identified **MAC bearer voice-channel activity** as the dominant discriminative signal via **feature importance analysis**, enabling lightweight real-time detection strategies
- Research 2. Insecurity of Lost/Stolen Phone Reporting Services** Jan. 2024 - Feb. 2026
- Demonstrated how lost/stolen reporting services can be exploited to launch **remote, silent device-DoS attacks against arbitrary users and devices** (e.g., *smartphones, tablets, IoT*)
  - Discovered **6 flaws** across device and network domains by auditing *3GPP/GSMA protocols* and analyzing the behavior of commercial reporting services through end-to-end security testing on major U.S. carriers (AT&T, Verizon, T-Mobile) and vendors (Apple, Samsung)

- Led **full-cycle development** of an end-to-end 5G SDR testbed (C/C++, *srsRAN*, *free5GC*), from architecture design through implementation, testing, and deployment of standard-compliant security patches

### Research 3. Insecure Designs of Cellular Emergency Services

Aug. 2021 - Dec. 2023

- Conducted **systematic threat analysis** of cellular emergency services, uncovering **4 critical attack vectors** that enable anonymous users to hijack highest-priority network access and disrupt 911 services; designed detection and mitigation mechanisms subsequently **adopted by AT&T**
- Built an **SDR-based 4G/5G network testbed** (*srsRAN*, *USRP*, *Open IMS Core*) and engineered a **custom emergency UE** atop *srsUE* (C/C++) to systematically benchmark topology resilience and validate standard-compliant defenses
- Developed an **Android testing toolkit** (*Emerg-Blocker*)[\[Github Link\]](#), demonstrating hands-on experience with SIP/IMS, RRC procedures, Android low-level controls, and **Qualcomm modem diagnostic trace analysis**
- Awarded **ACM MobiCom'22 Best Community Paper - Runner Up**, **AT&T Security Award**, **ACM SigMobile'24 Research Highlights**

### Research 4. M911-Verifier: Emerg-specific Model Checking Tool

Dec. 2023 - Jun. 2024

- Identified that flawed network selection can delay emergency calls even in strong-signal areas, with **90%** taking **over 2 minutes**, as well as causing call initiation failures and call drops
- Applied **model-checking techniques** to audit cellular emergency services in dynamic mobility scenarios, uncovering **11 counterintuitive logic flaws** in 4G/5G designs
- Architected an **Android application** (*Emerg-Dialer*)[\[Github Link\]](#) utilizing Telephony APIs for dynamic PLMN selection and VoWiFi-based routing, demonstrating system proficiency across SIM/eSIM management, roaming policies, and **IMS/VoWiFi network debugging**

### Research 5. Mobile Operating System Security

Aug. 2023 - Mar. 2025

- Discovered **4 vulnerabilities** across *baseband architectures*, *the Android Linux kernel*, and *IMS signaling routing* via **static and dynamic analysis** of attack surfaces, enabling traffic hijacking; recognized as **high-severity** by the **Google Bug Bounty Program**
- Designed and deployed a **cryptographic defense system** on an SDR testbed (*srsRAN*, *open5GS*) by modifying 5G core network functions (*AMF*, *PCF*, *gNB*) to integrate DHKE-based key exchange and MAC-based authentication for real-time IMS media session protection

## MEDIA COVERAGE

---

- **EurekAlert** (2025). "MSU researchers identify why 911 calls are delayed, failed, or dropped". March 27.
- **MSU Today** (2025). "MSU researchers identify why 911 calls are delayed, failed, or dropped". March 27.
- **ISSSource** (2023). "Boosting Security For Cellular Emergency Calls". May 1.
- **Homeland Security News Wire** (2023). "Making Emergency Calls More Secure". April 27.
- **Wood TV** (2023). "MSU researchers get \$1.2M to improve 911 call security". April 27.
- **EurekAlert** (2023). "MSU researchers to lead security improvements for cellular 911 calls with \$1.2M NSF grant". April 27.
- **Wilx TV** (2023). "MSU researchers awarded \$1.2 million grant to fix 911 cyber security issues". April 25.
- **MSU Today** (2023). "Making emergency calls more secure". April 24.

PROFESSIONAL SERVICES

---

External Reviewer

|                                                                                 |      |
|---------------------------------------------------------------------------------|------|
| • IEEE International Conference on Computer Communications (INFOCOM)            | 2025 |
| • IEEE International Conference on Computer Communications (INFOCOM)            | 2024 |
| • IEEE Military Communications Conference (MILCOM)                              | 2024 |
| • IEEE Communications Standards Magazine (COMSTD)                               | 2024 |
| • IEEE International Conference on Computer Communications (INFOCOM)            | 2023 |
| • IEEE International Conference on Network Protocols (ICNP)                     | 2023 |
| • IEEE International Conference on Computer Communications and Networks (ICCCN) | 2023 |
| • IFIP Networking Conference (IFIP)                                             | 2023 |
| • IEEE Transactions on Dependable and Secure Computing (TDSC)                   | 2023 |
| • IEEE Conference on Communications and Network Security (CNS)                  | 2022 |

TEACHING EXPERIENCE

---

|                                                                           |               |
|---------------------------------------------------------------------------|---------------|
| <b>CSE 410: Operating Systems</b>                                         | 2022-2023     |
| Undergraduate Course, Department of Computer Science and Engineering, MSU |               |
| Teaching Assistant                                                        |               |
| <b>CSE 425: Introduction to Computer Security</b>                         | (Spring) 2026 |
| Undergraduate Course, Department of Computer Science and Engineering, MSU |               |
| Teaching Assistant                                                        |               |
| <b>CSE 825: Computer and Network Security</b>                             | (Fall) 2025   |
| Graduate Course, Department of Computer Science and Engineering, MSU      |               |
| Teaching Assistant                                                        |               |

ACADEMIC MENTORING

---

|                                                                        |                         |
|------------------------------------------------------------------------|-------------------------|
| <b>Ph.D. Students (3)</b>                                              |                         |
| • Jared Singh                                                          | Fall 2024 - Present     |
| • Yu-An Chen                                                           | Fall 2024 - Present     |
| • Moyan Lyu                                                            | Fall 2024 - Present     |
| <b>Undergraduate Students(4)</b>                                       |                         |
| • Nishit Nagpal                                                        | Fall 2025 - Present     |
| • Jared Singh                                                          | Spring 2024 - Fall 2024 |
| Current position: Ph.D. student at Michigan State University (MSU)     |                         |
| • Felipe Marques Allevato                                              | Fall 2022 - Spring 2025 |
| Current position: Security Analyst at Microsoft in Redmond, Washington |                         |
| • Lukas Richters                                                       | Fall 2021 - Spring 2022 |
| Current position: Software Engineer at KLA in Ann Arbor, Michigan      |                         |

SKILLS

---

- **Security & Threat Research:** Vulnerability Research, Threat Analysis, Attack Modeling, Static/Dynamic Analysis, Applied Cryptography, Reverse Engineering, Responsible Disclosure (CVD), Model Checking
- **Programming & ML:** Python, C/C++, TensorFlow, PyTorch, XGBoost, LLM/RAG, Prompt Engineering, Shell Scripting, Raw-Socket Programming, Git, CI/CD
- **Systems & Platforms:** Linux, Android (Internals/ADB), srsRAN, open5GS, free5GC, USRP SDR, Wireshark, QXDM, OpenSSL, 3GPP/GSMA, 4G/5G, SIP/IMS

## REFERENCES

---

### **Guan-Hua Tu, Ph.D.**

Associate Professor (Advisor)  
Computer Science and Engineering Department  
Michigan State University  
Email: ghtu@msu.edu

### **Chi-Yu Li, Ph.D.**

Professor  
Computer Science Department  
National Yang Ming Chiao Tung University  
Email: chiyuli@cs.nycu.edu.tw

### **Songwu Lu, Ph.D.**

IEEE and ACM Fellow, Professor  
Computer Science Department  
University of California, Los Angeles  
Email: slu@cs.ucla.edu

### **Chunyi Peng, Ph.D.**

Associate Professor  
Department of Computer Science  
Purdue University  
Email: chunyi@purdue.edu